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Inventor(s)	Sheikholeslami; Sepehr et al.

Electromagnetic gripping device

Abstract

A system and method to provide a device in which the gripping action is achieved by a compliant membrane manipulated by electromagnetic forces. The gripping force is best suited for delicate objects, as it gently applies the gripping force necessary for displacement. This is accomplished through a chamber, a membrane, a plunger attached to the membrane, and a solenoid configured to manipulate the plunger, and thus, the membrane.

Inventors: Sheikholeslami; Sepehr (Vancouver, CA), Menon; Carlo (Zurich, CH)

Applicant: Nexera Robotics Corp. (North Vancouver, CA)

Family ID: 71842391

Assignee: NEXERA ROBOTICS CORP. (North Vancouver, CA)

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Primary Examiner: Vu; Stephen A

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The application claims priority and the benefit of U.S. Provisional Patent Application Ser. No. 62/799,870, entitled “MAGNETIC GRIPPER”, filed on Feb. 1, 2019 and PCT Patent Application Serial No. PCT/US2020/016463, entitled “ELECTROMAGNETIC GRIPPING DEVICE”, filed on Feb. 4, 2020 and U.S. patent application Ser. No. 17/427,912, entitled “ELECTROMAGNETIC GRIPPING DEVICE”, filed on Aug. 2, 2021, the disclosures of which are incorporated herein by reference in their entirety. This application is also a division of U.S. patent application Ser. No. 17/427,912 entitled “ELECTROMAGNETIC GRIPPING DEVICE”, filed on Aug. 2, 2021.

FIELD OF INVENTION

(1) This invention relates generally to an electromagnetic gripping device, and more particularly, to an end effector for robotic manipulators for grasping and moving objects using electromagnetic forces.

DISCUSSION OF RELATED ART

(2) In the world of automated manufacturing there are hundreds of different automated systems performing repetitive jobs. Robotic manipulators and robotic gripping devices are among the most desirable automated tasks where movement of delicate objects is required. Traditionally, robotic gripping devices such as flexible grippers and rigid grippers are developed with different actuation systems, and had their own advantages and disadvantages. With the need to pick different objects comes the need for flexible gripping devices.

(3) Flexible grippers are designed to handle objects in areas where rigid grippers are not best suited. In order to achieve the goal for more flexible grippers, numerous soft grippers have been designed which can pick and place objects with different shapes and weights. Some of these grippers need a vacuum pump or air compressor to actuate. For example, some variants incorporate suction cups and vacuums to lift objects through suction force. Other gripping devices are filled with granular material covered with an elastomer, where the elastomer forms around the object being lifted, and the air inside is evacuated to make it rigid and pick up the object.

(4) While there are different soft grippers in the art, they are limited in many aspects. The need

therefore exists for soft grippers to handle delicate objects without compromising speed and reliability. For example, there is the need to pick and place tomatoes in factories to put them one by one in an organized box. Furthermore, there is also a need for self-contained grippers that can pick up relatively heavier objects without the need for costly vacuum system in order to operate while not sacrificing speed of operation. The present invention satisfies these needs.

SUMMARY OF THE INVENTION

(5) The present invention will provide an electromagnetic gripping device adapted to handle delicate objects without compromising speed and reliability. Furthermore, the present invention is self-contained and adapted to pick up relatively heavier objects without the need for additional systems and without increasing operating speed. This is accomplished through a deformable membrane, a plunger attached to the membrane, and a solenoid configured to manipulate plunger, and thus the membrane, about an object. These elements work in conjunction to provide grip onto an object using electromagnetic force, or more specifically, by deforming the membrane such that it provides the grip necessary to lift and displace the object.

(6) These and other objectives of the present invention will become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiments. It is to be understood that the foregoing general description and the following detailed description are exemplary, and are intended to provide further explanation of the invention as claimed.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front cross-sectional view of the present invention with a single plunger;
- (2) FIG. 2 is a front cross-sectional view of the present invention showing a plurality of plungers;
- (3) FIG. 3 is a front cross-sectional view of the present invention showing a plurality of stacked plungers;
- (4) FIG. 4 is a front cross-sectional view of the present invention showing the membrane positioned within the solenoid;
- (5) FIG. 5 is a front cross-sectional view of the present invention showing an open-ended chamber;
- (6) FIG. 6 is a front cross-sectional view of the present invention showing an open-ended chamber with a conoidal-shaped chamber;
- (7) FIG. 7 is a front cross-sectional view of the present invention showing a conoidal-shaped chamber attached to a pressurized line;
- (8) FIG. 8 is a front cross-sectional view of the present invention showing truncated cone;
- (9) FIG. 9 is a front cross-sectional view of the present invention showing membrane with variable thickness;
- (10) FIG. 10 is a front cross-sectional view of the present invention showing latch;
- (11) FIG. 11 is a front cross-sectional view of the present invention showing a plurality of sensors;
- (12) FIG. 12 is a front cross-sectional view of the present invention showing a cooling system adjacent to the solenoid;
- (13) FIG. 13 is a front cross-sectional view of the present invention showing a cooling system positioned between two the solenoids;
- (14) FIG. 14 is a front cross-sectional view of the present invention showing a cooling plate adjacent to the solenoid;
- (15) FIG. 15 is a front cross-sectional view of the present invention showing a plurality of solenoids;
- (16) FIG. 16 is a front cross-sectional view of the present invention showing a solenoid in a different configuration;
- (17) FIG. 17 is a front cross-sectional view of the present invention showing an object being

gripped and suction created between the object and the membrane;

(18) FIG. **18** is a front perspective view of the present invention;

(19) FIG. **19** is a flow diagram showing the steps of the present invention;

(20) FIG. **20** is a flow diagram showing the steps of the present invention;

(21) FIG. **21** is a flow diagram showing the steps of the present invention;

(22) FIG. **22** is a flow diagram showing the steps of the present invention;

(23) FIG. **23** is a cross-sectional view of the present invention using a rod to guide the plunger;

(24) FIG. **24** is a cross-sectional view of the present invention having an open end and using a rod to guide the plunger;

(25) FIG. **25** is a cross-sectional view of the present invention showing a large plunger attached to a loose membrane;

(26) FIG. **26** is a cross-sectional view of the present invention showing a large plunger attached to a loose membrane with a guide groove;

(27) FIG. **27** is a cross-sectional view of the present invention gripping a random shape object showing the suction created between the object and the membrane and the formation of the membrane around the object;

(28) FIG. **28** is a cross-sectional view of the present invention where the plunger is in the form of small-scale particles embedded in the membrane; and

(29) FIG. **29** is a cross-sectional view of the present invention where a connector is shown.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(30) Illustrative embodiments of the invention are described below. The following explanation provides specific details for a thorough understanding of and enabling description for these embodiments. One skilled in the art will understand that the invention may be practiced without such details. In other instances, well-known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments.

(31) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising,” and the like are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” Words using the singular or plural number also include the plural or singular number respectively. Additionally, the words “herein”, “above”, “below” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. When the claims use the word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list and any combination of the items in the list.

(32) The present invention comprises a chamber **20**, a solenoid **21** coiled about the chamber **20**, a deformable membrane **31**, and a plunger **41** attached to the deformable membrane **31**. These components are described in various configurations and work in conjunction to provide grip onto an object **91**, or more specifically, by deforming the membrane **30** through electromagnetic forces such that the membrane **31** creates the negative pressure (suction) and/or other forces necessary to lift and displace an object **91**. The electromagnetic interaction between the solenoid **21** and the plunger **41** induces movement of the membrane **31**, and as such, deformation of the membrane **31** around the object **91**. Further movement of the object **91**, depending on the roughness and porosity of the object **91**, can create a suction pocket between the object **91** and the membrane **31**, further increasing the gripping potential of the present invention.

(33) In the preferred embodiment, the chamber **20** comprises a structural support for the device and is generally cylindrical. In the preferred embodiment, the chamber **20** will have at least one open end. Alternatively, the chamber **20** may comprise a closed end, forming a sealed chamber along with the membrane **31**. The chamber **20** may be of any suitable shape, such as cylindrical, rectangular, or conoidal. In an alternative embodiment, a plurality of chambers **20** may be connected and work in conjunction to provide the forces necessary to move the plunger **41**. The

chamber **20** comprises a durable and resilient material, such as aluminum or other metal alloy. Alternatively, the chamber **20** may be made of a durable, resilient, and non-magnetic material. In a further alternative embodiment, a plurality of sensors are incorporated to detect the position of the membrane **31**, plunger **41**, the object **91**, the temperature of the device, the pressure of the device, and other environmental information.

(34) The solenoid **21** is positioned within or around the perimeter of the chamber **20** and comprises conductive wire wrapped around the chamber **20** generally in the shape of a spiraled coil and adapted to generate a magnetic field when current is applied to the solenoid **21**. The magnetic field generated by the solenoid **21** is adapted to interact with the plunger **41**, which is positioned adjacent to or within the solenoid **21**. The strength of the magnetic field varies by application, and as such, the design criteria of the present invention may change depending on the application. More specifically, the strength of the magnetic field is modified by increasing or decreasing the number of wire turns the solenoid **21** is wrapped around the chamber **20**, by increasing or decreasing the diameter and length of the wire in the solenoid **21**, by increasing the current applied to the solenoid **21**, or by changing the material used in the solenoid **21**. Furthermore, the direction of the magnetic field can be switched by changing the polarities of power applied to the solenoid **21**.

(35) The solenoid **21** is conductive and its conductivity is dependent on the material used. The solenoid **21** may be made of copper, silver, gold, or other suitable conductive material depending on the power, efficiency, and cost needs of the application. Furthermore, the solenoid **21** can be a stretchable and flexible (e.g. conductive polymer). In an alternative embodiment, the present invention may comprise a plurality of solenoids **21** in series, parallel, or other configuration adapted to work in conjunction to create a single magnetic field to interact with the plunger **41**. For example, a plurality of solenoids **21** may be stacked, each having an independent set of terminals, that may be powered individually or may be powered in series or in parallel. Furthermore, the plurality of solenoids may have different wire thicknesses, each requiring different current, and may further be wound adjacent to one another to provide a more controlled electromagnetic field. While a generally spiral shape is shown in the figures, the overall shape of the solenoid **21** can be in the form of cylindrical, conical, toroidal, helical or other geometry where the magnetic field can be created to interact with the plunger **41**.

(36) The membrane **31** comprises a flexible and durable substrate capable of providing substantial grip and is positioned adjacent to the solenoid **21**. The membrane **31** is adapted to come into direct contact with the object **91** and deform around a part of or the within the chamber **20**, although the movement of the membrane **31** within the chamber **20** can vary depending on the application. Suitable membrane **31** materials include latex, silicone, polyurethane, and other polymers or flexible materials. The membrane **31** can also be made in forms of circular, rectangular, cylindrical with a closed end, and any other shape depending on the shape of the solenoid **21** and plunger **41**. Furthermore, the membrane **21** can have constant or variable thickness. Lastly, the surface of the membrane **31** can be smooth to better comply with smooth objects, or structured for objects with a rougher surface or where the structured surface can dictate how and where the membrane **31** will fold to maximize grasping and suction forces.

(37) The membrane **31** will provide direct gripping force to the object **91** through the pressure acted by the membrane **31** as a result of deformation in proximity of the object **91**. The membrane **31** will also apply other gripping forces, comprising friction, suction, dry adhesion, electro-adhesion, magnetic adhesion, negative pressure, or other forms of attractive force provided by the mechanical, chemical, or nuclear properties of the membrane **31** and the object **91**. Dry adhesion is based on van der Waals forces created at the surface of the membrane **31**. These van der Waals forces, based on electric dipole interactions, can be enhanced by means of electro dry adhesion. As such, electro dry adhesion can increase the preload of the membrane by exerting an electric force toward the object to be lifted. For this purpose, electrodes can be embedded into the membrane **31** and charged accordingly to increase the preload and to reduce the pressure applied on the object by

the membrane.

(38) More specifically, the membrane **31** may be specifically designed to provide attractive and/or repellant forces independent of its deformation. Here, the membrane **31** will further comprise a micro-featured surface adapted to provide dry adhesion, and at least one conductive material adapted to provide electrostatic adhesion. A voltage is applied to the conductive material(s), providing an attractive force to self-preload said micro-featured surface, a resistive force to self-peel said micro-featured surface, or alternating said voltage to self-clean said micro-featured surface. The membrane **31** may comprise a conductive polymer, rubber or silicone rubber or a composite polymer, rubber, or silicone rubber material with embedded conductive particles such as carbon black, carbon nanotubes, silver particles, graphene, graphene, graphite, or other conductive materials, and one electrode.

(39) The plunger **41** comprises a magnetic or ferromagnetic material adapted to react to the magnetic field generated by the solenoid **21**. The plunger **41** is directly attached to the membrane **31**, and as such, when the plunger **41** moves, so does the membrane **31**. More specifically, the power of the magnetic force applied to the plunger **41** from the solenoid **21** will cause the membrane **31** to move into a controlled position, and as the power and magnetic force change, so does the position of the plunger **41**, which in turn maintains the deformation of the membrane **31** as the object is displaced through the controlled position of the plunger **41**. For example, if an object is positioned below the membrane **31**, the current sent through the solenoid **21** may decrease or reverse, enabling the membrane **31** to lower and come into direct contact with the object **91**, after which the current will increase, raising the membrane **31** and causing friction, suction, and other forces to grip and lift the object **91**.

(40) In the preferred embodiment, the plunger **41** is positioned at the center of the membrane **31**. Alternatively, the plunger **41** can be positioned elsewhere on the membrane **31** based on the application and it does not necessarily need to be in the center. Although various plunger **41** shapes are provided in the drawings, the plunger **41** can be of any suitable shape such as rectangular, cylindrical, or other shapes so long as it interacts with the magnetic field provided by the solenoid **21**. More specifically, the size and shape of the plunger **41**, along with the material used in the plunger **41** and its position within the device, directly influence the plunger's **41** interaction with the magnetic field, and as such, the size, shape, position, and material of the plunger are all selected based on the requirements of the application. In an alternative embodiment, a plurality of plungers **41** are attached to the membrane **31**. In a further alternative embodiment, the plunger **41** is isolated from the membrane **31**. In yet a further alternative embodiment, the plunger **41** is doped or otherwise combined into the membrane **31** such that the membrane **31** is directly manipulated by the solenoid **21** without the plunger **41**. In yet a further alternative embodiment, the plunger **41** is hollow and filled with an aggregate magnetic material.

(41) FIG. **19** is a flow diagram describing the grasping method **201**. First, the solenoid **21** is activated **221**. As a result, the plunger **41** and/or a membrane **31** is activated **222** because of the solenoid's **21** magnetic field. Hence, the object **91** is grasped **223**.

(42) In another method **202** as shown in FIG. **20**, the membrane **31** is in direct contact with an object **91** in the first step **231**. In the second step **232**, the membrane's **31** grasping forces, that were explained earlier such as friction or electro-adhesion, provide gripping force to the object **91**. Then the solenoid **21** is activated to increase or generate additional grasping forces **233**, such as friction and suction. Hence the object **91** is grasped **234**.

(43) The figures illustrate various embodiments that may be used when practicing the present invention. In FIG. **1**, a solenoid **21**, a plunger **41**, and a membrane **31** are used to grasp an object **91**. Shown in FIG. **21** is a flow diagram **203** of a method of gripping an object. In the first step **241**, the object **91** moves, and then it contacts the gripping device as shown in step **242**. The solenoid **21** is then activated **243**, and the plunger **41** is moved further away from the object **91** due to the interaction with the solenoid **21**. Based on the roughness and porosity of the object, a seal may be

created between the object **91** and the membrane **31** through the movement of the plunger **41** by the solenoid **21**, creating a suction pocket. Through the suction pocket and/or other forces applied to the object **91** by the gripping device **100**, the membrane **31** can grasp **244** the object **91**.

(44) In an alternative embodiment as shown in FIG. **22**, a flow diagram **204** of a method of gripping an object **91** is shown where the solenoid **21** is moved **251** towards the object **91**, and, as in step **252**, the gripping device contacts the object **91**. Then the solenoid **21** gets activated **253** and the gripping forces by the membrane **31** and/or the suction pocket allows grasping **254** the object **91**.

(45) In a further alternative embodiment, the solenoid **21** induces a deformation of the membrane via the plunger **41**. In another embodiment, the plunger **41** is positioned in the axis of the solenoid **21**.

(46) In a further alternative embodiment, as shown in FIGS. **1** and **18**, the gripping device **100** comprises a solenoid **21**, a membrane **31**, and a plunger **41**. The plunger **41** is positioned near or in proximity of the center of the membrane **31**. The solenoid **21** and the membrane **31** could be connected together via a connector having different shapes and materials. Such a connector could have a protruded or retracted shape. This could be used to facilitate the membrane **31** to conform to the object to be grasped. FIG. **29** shows an example of a gripping device **143**, where a connector **171** is attached to the membrane **36**. Such a connector **171** could be fixed to the solenoid **23** or the chamber **53** or other parts of the gripper.

(47) In more detail, and still referring to FIGS. **1** and **18**, the solenoid **21** can be designed for picking specific objects, hence its size could be designed based on the object being displaced. The solenoid **21** in this instance is in the shape of a cylinder. Depending on the design criteria, the wire can be of different diameters with different number of turns hence defining the solenoid's **21** inner diameter, outer diameter, and length. The membrane **21** is attached adjacent to the solenoid **21** on its edges by means of any bonding material such as epoxy, or the membrane **21** can be cured when it is already in contact with solenoid **21**. The membrane **21** shown in FIG. **1** is attached to one end of the solenoid **21**, but it does not necessarily need to be connected to the end of the solenoid **21**; in other words, it can also be connected in the middle of the solenoid **21**, or even positioned between 2 solenoids. The plunger **41** in this embodiment is attached to the membrane **31** using any material that can bond to the membrane **31**. The plunger **41** can also be embedded in the membrane **31**. The proportionality of the plunger **41** with respect to the solenoid **21** is important and it needs to be chosen depending on size and weight of objects being picked.

(48) Referring now to FIG. **2**, there is shown another version of the gripping device **101**, where the solenoid **21**, the membrane **31**, and a plunger **42**, which is in the form of small particles attached to the membrane **31**. The plunger **42** can be also embedded into the membrane **31**. For instance, the membrane **31** could consist of a polymer doped with ferromagnetic or magnetic particles. An illustration is provided in FIG. **28** showing the membrane **35** embedded with particles **47**.

(49) Referring now to FIG. **3**, there is shown another version of the gripping device **108** where the solenoid **23**, and the membrane **31** are present with a plunger **43** which is a stack of single plungers. Any number of pieces can be stacked together to form plunger **43**, hence change its length as needed. For instance, if the plunger **43** is magnetized then by adding more to the stack, we can increase the final gripping force since the plunger **43** is attracted or repelled by the solenoid **23**. Therefore, gripping strength of gripping device **108** can be increased or decreased based on the object it is picking without doing major changes in the design.

(50) In another embodiment as shown in FIG. **4**, the gripping device **109** has a solenoid **24** connected to a solenoid **25** either in series or parallel depending on design requirements. In this embodiment, the membrane **31** is located in between the two solenoids **24** and **25**. The location of the membrane **31** helps positioning the plunger **41** to a desired location to better grip the object **91**. In general, the size and location of the plunger **41** with respect to the solenoid **21** is an important design factor to reach required gripping strength where it is needed. The magnetic field of the

solenoid **24** and **25** interacting with the plunger **41** changes depending on location of the plunger **41** with respect to the overall solenoid **21**.

(51) In another embodiment as shown in FIG. 5, the gripping device **102** has a chamber **20** where the solenoid **21** is wrapped around. This chamber **20** can be of different forms to better help with the gripping process. In one instance, the chamber **51** is in the form of a half toroid attached to a cylinder as shown in FIG. 5. In fact, different shapes of the gripping device can facilitate the gripping of an object by further covering more surface area of the object with a membrane **32**, and also helps direct the object to the axis of the solenoid **21**. The chamber **51** can also change the position of the plunger **41** with respect to the solenoid **21** by having the extra half toroid on the bottom, hence displacing the membrane **32**. For example, it can place the plunger **41** further away from the solenoid **21** to create stronger magnetic fields in gripping positions where more force is needed. In other word, after the object is closer to the solenoid **21**, then the plunger **41** is positioned in a stronger magnetic field in the solenoid **21**.

(52) In another embodiment as shown in FIG. 6, there is a gripper **103** with a conical shape chamber **52**, with a solenoid **22** wrapped around it in the same form. This conical form of the chamber **52** can have different advantages. For instance, it can further increase the gripping force by reducing the solenoid's **22** inner diameter as the object is grasped and further moved up the axis of the solenoid **22**. Depending on the relative size of the plunger **41** with respect to the solenoid **22**, the magnetic force affecting the plunger **41** can be increased since the size of plunger **41**, and solenoid **22** are not changing in this embodiment, but the distance from the plunger **41** to the solenoid **22** is reduced as the plunger **41** further moves up the solenoid's **22** axis.

(53) In another embodiment as shown in FIG. 7, the gripping device **104** has the solenoid **22** wrapped around a chamber **53** where one end of the chamber **53** is attached to a pressurizes gas or fluid line. In this embodiment the plunger **41** can be pulled up or attracted by the solenoid **21** when gripping an object and the pressurized gas or fluid can push the plunger **41** away to release the grasped object. Therefore, the object can have a tighter and stronger fit in the chamber **53**, since the force of the pressurized gas or liquid is strong enough to repel the plunger **41** depending on the application.

(54) In another embodiment as shown in FIG. 8, the gripping device **105** has a solenoid **23**, a membrane **32**, and the plunger **41**. In this embodiment a truncated membrane **32** is used. This form of membrane **32** allows more flexibility in picking up the object. For instance, the membrane **32** surface area is bigger in compare to previous embodiments. In other words, the membrane **32** will be less stretched when the object **91** is grasped and moved inside the solenoid **21**. Therefore, less holding force is needed to hold the object **91** or less magnetic force is needed to hold the plunger **41** up in the solenoid **21**. Most of the magnetic force in this embodiment is used to hold the plunger **41** up in the solenoid **23** and consequently hold the object **91** in grasp position, and ideally zero force is used to compensate the stress caused by the stretch in the membrane **32**.

(55) In another embodiment as shown in FIG. 9, the gripping device **106** consists of the solenoid **23**, a membrane **33**, and the plunger **41**. In this instance, the membrane **33** has a variable thickness. It is used to better control the stretch ratio in different areas of the membrane **33**. In other words, thinner areas will stretch more than thicker parts with the same stress applied to the membrane **33**. This idea also helps to have a better suction to grip the object **91** as shown in FIG. 17, which will be explained in more detail in another embodiment.

(56) In another embodiment as shown in FIG. 10, the gripping device **107** consists of the solenoid **23**, the membrane **31**, the plunger **41**, a chamber **54**, and a latch **61**. In this configuration, the solenoid **23** is wrapped around the chamber **54**, and the chamber **54** has a bracket on one end holding the latch **61**. This latch **61** can be in the form of any device that can attract the plunger **41**. For instance, it can be a solid piece made of ferromagnetic or magnetic material, or it can even be a latching mechanism which can be activated or deactivated to attract the plunger **41**. This latch **61** can decrease the power consumption of the gripping device **107**. When an object is being grasped

by the gripping device **107**, the latch can also attract to hold the plunger **41** further up in the solenoid **23**, hence reduce the magnetic force needed by the solenoid **23**.

(57) In an alternative embodiment, the plunger **41** can be pulled by the solenoid **23** and then the solenoid power is disconnected and the latch **61** holds the plunger **41** while gripping. Then in order to release the plunger **41** and consequently the object **91**, the polarity of the solenoid **23** can be reversed to push the object **91** out and release it. In this example the plunger **41** can be magnetic, therefore by reversing the solenoid **23** polarity it can be pushed or pulled in the solenoid **23**.

(58) In another embodiment as shown in FIG. **11**, the gripping device **110** has the solenoid **23** wrapped around a chamber **55** which incorporates a set of sensors **71**. For instance, these sensors **71** can be used to detect the object **91** being grasped, measure environment attributes such as pressure and temperature, or act as a trigger for another device. These sensors can be based on infrared light or other principles.

(59) The coiled wire has a resistance and produces magnetic field; hence it can become hot, especially when it is continuously active. Therefore, a cooling device can be used to cool down the solenoid's temperature for better performance and prolonged use. FIGS. **12-14** illustrate embodiments where a cooling device is incorporated into the gripping device to reduce the solenoid temperature. For instance, FIG. **12** shows a gripping device **111** which has a cooling device **81** wrapped around the solenoid **23**. The cooling device **81** can be in the form of hollow tubes and pass cooling liquid or gas to take away the increased temperature from the solenoid **23**. The cooling device **81** can also be made of a conductive material and also act as a second solenoid and be connected to the solenoid **23** either in series to in parallel.

(60) Now referring to FIG. **13**, the gripping device **112** has also another solenoid **26** wrapped around the cooling device **81**. In this embodiment, the cooling device **81** not only cools down both solenoid **23** and **26**, but also can act as a solenoid itself and be connected in series or parallel with solenoid **23** and **26**. In another embodiment as shown in FIG. **14**, a gripping device **113** has a cooling device **82** which is a cylinder covering the solenoid **23**. In this embodiment the cooling device **82** has a large surface area, and it can dissipate by the means of liquid or air cooling. For instance, for a better air cool, the cooling device **82** can also have fins around it and a fan on top to move cool air in between the fins and remove the heat from the other end.

(61) In another embodiment as shown in FIG. **15**, the gripping device **114** has three solenoid **26-28**. A chamber **56** is constructed to have the solenoids **26-28** wrapped around its branches. For instance, solenoids **26-28** can be used to better control the movement direction of the plunger **41**. Depending on design requirements, solenoids **26-28** can be completely different from each other. They can also be connected to a controller individually or connected together in series or parallel. While FIG. **15** includes three solenoids, the number could be higher or lower and their position can be arranged differently depending on the application.

(62) In another embodiment as shown in FIG. **16**, the gripping device **115** is shown which has a plunger **44** around a solenoid **29**. In this embodiment the magnetic field around outside the solenoid **29** interacts with the plunger **44** to induce the membrane **31** to provide the gripping or attractive force.

(63) In another embodiment as shown in FIG. **17**, the gripping device **116** is shown gripping an object **91**. In this embodiment, the solenoid **23** has pulled the plunger **41** further higher than the object **91**. As a result, a suction is created between the membrane **31** and the object **91**. Therefore, not only the surface texture and friction between the membrane **31** and the object **91** is gripping the object **91**, but also the suction pocket created between the object **91** and the membrane **31** increases the gripping force. The object **91** does not need to be necessary around the object, but it can be of different shapes. For instance, if the object **91** is flat, then the suction created has a big impact in the gripping strength. The suction in this embodiment is caused by the ratio between the initial volume and the final volume of the pocket between the object **91** and the membrane **31**. The initial volume is considered to be the volume when the membrane **31** is just touching the object **91**. Such

an initial volume can be nil. The final volume is the volume of the suction pocket shown in FIG. 17.

(64) In more detail, and still referring to FIG. 17, the suction strength can also be increased or decreased by changing design attributes. For instance, the solenoid 21 magnetization field can be decrease or increase to position the plunger 41 lower or higher respectively and hence decrease or increase the suction pocket created. In another instance, the membrane 33 can have variable thickness as shown in FIG. 9. One example is to increase the thickness of the membrane 33 in the part where the suction pocket is located. This extra thickness can cause the membrane 33 to stretch less in this part, and consequently will change the volume ratio creating the suction pocket. Therefore, the membrane 33 can have variable thickness, shaped as a truncated cone, flat circle, combination of them all, or any other shape. In other words, different shapes of the membrane 33 can have an effect on increasing and decreasing the suction pocket created.

(65) The advantages of the present invention include, without limitation, its simplicity, speed, versatility, size, and the ability to pick large variety of objects, including delicate objects, objects having different shapes and made of different materials and surface finish. It is a fast action gripper since its main actuation is based on a solenoid 21. This invention can also use a suction force to grip an object without the need for a separate vacuum line. The design also minimizes the amount of mechanical wear. Further the device can easily mount on an industrial robot and move in tight spaces.

(66) In another embodiment as shown in FIG. 23, another version of the gripping device 117 is shown. The gripper 117 has a rod 121 attached to the plunger 41 fixed to the membrane 31. The plunger 41 and the rod 121 can be one single element. The plunger 41 and the rod 121 could have different shapes and orientations. The gripper 117 could have a cover 131. The cover 131 could have a shape to limit the movement of the rod 121. For example, the cover 131 could have a groove to guide the rod 121 in the center of solenoid 21 axis. The cover 131 could be made of a single or multiple parts and materials. The cover 131 could be part of a chamber that seals the membrane 31 such that a gas or fluid could be contained within the chamber and a membrane 31.

(67) In another embodiment as shown in FIG. 24, gripper 118 is shown, where the cover 132 has holes to expose the air inside coil 21 to surrounding pressure. Holes could have different shape or could be substituted by a porous material.

(68) In another embodiment as shown in FIG. 25, gripper 119 is shown. The plunger 45 can be in different size and shapes. For instance, the plunger 45 shown in FIG. 25, has an elongated shape. It should be noted that the membrane 32 can be loose as shown in FIG. 25. Such a loose membrane 32 could farther facilitate the membrane itself to conform to the object's shape. As shown in FIG. 26, the gripper 141 can have a cover 131. The plunger 45 can move within the groove limits of the cover 131.

(69) In another embodiment as shown in FIG. 27, the gripper 142 is shown gripping an object 92. Object 92 can be of any shape and orientation. As shown in FIG. 27, the membrane 34 is formed around the object 92 and also a suction cavity 151 is formed between the object 92 and the membrane 34. Formation of the membrane 34 around the object 92 and the suction cavity 151 helps with gripping the object 92. As previously mentioned, the plunger 41 could have different shapes. FIG. 27 shows an example of a plunger 46 which has a conical shape.

(70) In a broad embodiment, the present invention consists of at least a solenoid 21, at least a plunger 41, and at least a membrane 31. The interaction among the solenoid 21, the plunger 41, and the membrane 31 induces the membrane 31 to provide a gripping or attractive force.

(71) In a broad embodiment, the present invention consists of a method of grasping objects 91 using at least a solenoid 21, at least a plunger 41, and at least a membrane 31.

(72) While the above description contains specific details regarding certain elements, sizes, and other teachings, it is understood that embodiments of the invention or any combination of them may be practiced without these specific details. Specifically, although certain materials and shapes

are designated in the above embodiments, any suitable materials or shapes may be used. These details should not be construed as limitations on the scope of any embodiment, but merely as exemplifications of the presently preferred embodiments. In other instances, well known structures, elements, and techniques have not been shown to clearly explain the details of the invention.

(73) The above detailed description of the embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed above or to the particular field of usage mentioned in this disclosure. While specific embodiments of, and examples for, the invention are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. Also, the teachings of the invention provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

(74) Changes can be made to the invention in light of the above “Detailed Description.” While the above description details certain embodiments of the invention and describes the best mode contemplated, no matter how detailed the above appears in text, the invention can be practiced in many ways. Therefore, implementation details may vary considerably while still being encompassed by the invention disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the invention should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the invention with which that terminology is associated.

(75) While certain aspects of the invention are presented below in certain claim forms, the inventor contemplates the various aspects of the invention in any number of claim forms. Accordingly, the inventor reserves the right to add additional claims after filing the application to pursue such additional claim forms for other aspects of the invention.

Claims

1. An electromagnetic gripping device, comprising: at least one solenoid made of conductive material, wherein said solenoid generates an electromagnetic field when an electric current is applied to it; at least one deformable membrane in the form of an open surface with one boundary; wherein said membrane can be actively deformed by changing the electromagnetic field generated by said solenoid; wherein said deformable membrane can be passively deformed by conforming to an object to be grasped upon contact with said object; wherein said deformation of said deformable membrane provides a gripping or attractive force when said deformable membrane is positioned adjacent to said object; wherein said deformable membrane is in direct contact with and conforms to said object when said gripping device is holding said object.
2. The gripping device of claim 1, further comprising: a plunger attached to said deformable membrane; wherein said plunger comprises ferromagnetic, ferrimagnetic, or magnetic material configured to interact with said electromagnetic field generated by said solenoid; wherein said interaction comprises changing said electric current applied to said solenoid, wherein said changes in said electric current change the strength and direction of said electromagnetic field, and wherein said changes in said electromagnetic field manipulate the movement and position of said plunger within said electromagnetic field; wherein said interaction between said plunger and said electromagnetic field generated by said solenoid actively deforms said deformable membrane.
3. The gripping device of claim 2, further comprising: a guiding structure to guide the movement of said plunger.
4. The gripping device of claim 2, further comprising: a latch configured to be able to attract or hold said plunger when said object is being grasp.
5. The gripping device of claim 2 configured for gripping an object, the method comprising the steps of: moving the gripping device towards the object to be grasped such that said deformable

membrane is in direct contact with said object; moving the gripping device further towards the object such that said deformable membrane deforms by conforming passively to said object to be grasped, exerting a grasping force, at least in the form of friction between said deformable membrane and said object to be grasped; applying an electric current to said solenoid to generate an electromagnetic field; and generating an electromagnetic field in said solenoid to displace said plunger; wherein displacement of said plunger solenoid actively deforms said deformable membrane; wherein deformation of said deformable membrane provides additional gripping forces, at least in the form of friction, to said object to be grasped.

6. The gripping device of claim 1, wherein said membrane further comprising ferromagnetic, ferrimagnetic, or magnetic material configured to interact with said electromagnetic field generated by said solenoid; wherein the interaction between said ferromagnetic, ferrimagnetic, or magnetic material in said membrane and said electromagnetic field generated by said solenoid actively deforms said deformable membrane.

7. The gripping device of claim 1, further comprising: a chamber with at least one open end, configured to provide structural support to the device; wherein said solenoid is positioned within or around said chamber; wherein said membrane is attached to an open end of said chamber.

8. The gripping device of claim 7, further comprising at least one sensor positioned within said chamber, configured to detect the pressure inside said chamber, the volume inside said chamber, the temperature of said solenoid, or the object.

9. The gripping device of claim 7, further comprising a plunger attached to said deformable membrane; wherein said plunger comprises ferromagnetic, ferrimagnetic, or magnetic material configured to interact with said electromagnetic field generated by said solenoid; wherein the interaction between said plunger and said electromagnetic field generated by said solenoid actively deforms said deformable membrane.

10. The gripping device of claim 9, further comprising at least one sensor positioned within said chamber, configured to detect the pressure inside said chamber, the volume inside said chamber, the temperature of said solenoid, the position of said plunger, or the object.

11. The gripping device of claim 9, wherein at least one opening of said chamber is connected to a pressurizes gas or fluid line.

12. The gripping device of claim 9, further configured for gripping an object, wherein said chamber has only one open end to which said deformable membrane is attached; and wherein said chamber and said deformable membrane together encloses a volume filled with a gas or a fluid.

13. The gripping device in claim 1, further comprising: a connector shaped to facilitate said deformable membrane to deform and conform to said object to be grasped; wherein said solenoid is connected to one end of said connector, and said deformable membrane is connected to the other end of said connector.

14. The gripping device of claim 1, further comprising: a chamber with at least one open end configured to provide structural support to the device; a connector configured to facilitate said deformable membrane to deform and conform to said object to be grasped; wherein said solenoid is positioned within or around said chamber; wherein an open end of said chamber is attached to one end of said connector; wherein said membrane is attached to the other end of said connector.

15. The gripping device of claim 1, wherein said membrane further comprises a micro-featured surface adapted to provide a gripping or attractive force in the form of dry adhesion.

16. The gripping device of claim 1, wherein said membrane further comprises at least one conductive material adapted to provide a gripping or attractive force in the form of electrostatic adhesion, wherein a voltage is applied to said at least one conductive material providing an attractive force to self-preload said membrane, a resistive force to self-peel said membrane, or alternating said voltage to self-clean said membrane.

17. The gripping device of claim 1, wherein said solenoid at least partially comprises hollow tubes; wherein coolant may flow through said hollow tubes, reducing the temperature of said solenoid.

18. The gripping device of claim 1 further configured for fabricating the deformable membrane and attaching it to the solenoid in the gripper device in claim 1, the method comprising the steps of: dipping a least a part of said solenoid into the liquid precursor of said deformable membrane; and curing said liquid precursor to form said deformable membrane, such that said deformable membrane is attached to said solenoid upon fabricated.

19. A method of gripping an object using a gripping device, the method comprising the steps of: moving the gripping device towards the object to be grasped such that a deformable membrane is in direct contact with said object; moving the gripping device further towards the object such that said deformable membrane deforms by conforming passively to said object to be grasped, exerting a grasping force, at least in a form of friction between said deformable membrane and said object to be grasped; applying an electric current to a solenoid to generate an electromagnetic field; said generation of electromagnetic field in said solenoid actively further deforms said deformable membrane; and said further deformation of said deformable membrane provides additional gripping forces, at least in the form of friction, to said object to be grasped.

20. A method of gripping an object using a gripping device, the method comprising the steps of: moving the gripping device towards the object to be grasped such that a deformable membrane is in direct contact with said object; moving the gripping device further towards the object such that said deformable membrane deforms by conforming passively to said object to be grasped, exerting a grasping force, at least in a form of friction between said deformable membrane and said object to be grasped; applying an electric current to a solenoid to generate an electromagnetic field; and activating said latch to attract and hold said plunger; wherein generation of electromagnetic field in said solenoid displaces said plunger; wherein displacement of said plunger solenoid actively deforms said deformable membrane; and wherein deformation of said deformable membrane provides additional gripping forces, at least in the form of friction, to said object to be grasped.
